

CRF: Clifford Algebra $\text{Cl}(3)$ Structure

Verification, Grade Conservation, and the Programme for Paper 2

Ougit Jarittum

April 2026 ORCID: [0009-0009-4451-6811](https://orcid.org/0009-0009-4451-6811) CC-BY-NC 4.0

Companion to CRF Paper v7 and CRF FormStructure. Verification session for the Clifford connection.

Abstract. We verify the claim that the $n = 8$ CRF mode space is functionally the Clifford algebra $\text{Cl}(3, 0)$. Seven independent features are established, ranging from grade counting to dynamical grade conservation in W_{ij} .

The connection is *not* a mere grade-counting coincidence: the W_{ij} coupling is a Clifford grade-0 scalar, the CRF ground state is the Clifford null state, the pseudoscalar $I_3 = e_x e_y e_z$ satisfies $I_3^2 = -1$ (providing the imaginary unit of quantum mechanics), and the bivectors generate the $\text{SU}(2)$ spin algebra.

The remaining question — what Clifford multiplication means for CRF Born-rule dynamics — is identified as the programme for the next paper.

Seven Evidence Points

(A) Grade structure: $1 + 3 + 3 + 1 = 8 = n$ (exact)

The Clifford algebra $\text{Cl}(3, 0)$ decomposes by grade:

Grade	Count $\binom{3}{k}$	CRF mode	Role
0	1	$\emptyset$	scalar
1	3	$\{x\}, \{y\}, \{z\}$	space
2	3	$\{xy\}, \{xz\}, \{yz\}$	gauge
3	1	$\{xyz\}$	pseudoscalar
Total	8	$= n$	

(B) $I_3^2 = -1$ (proved from Clifford axioms)

$$I_3^2 = -1$$

The unit pseudoscalar $I_3 = e_x e_y e_z$ satisfies:

$$I_3^2 = -1 \quad (\text{formula: } I_n^2 = (-1)^{n(n-1)/2}, n = 3)$$

Verified numerically via Pauli matrix representation ($I_3 = iI_2$, so $I_3^2 = -I_2$).

Consequence: $\text{Cl}(3, 0) \cong M_2(\mathbb{C})$ where $i = I_3$.

(C) I_3 in the center (proved)

$I_3 = iI_{2 \times 2}$ commutes with every 2×2 matrix. Therefore I_3 is in the center of $\text{Cl}(3, 0)$. Physically: the pseudoscalar mode $\{xyz\}$ provides a global $U(1)$ phase symmetry that commutes with all other modes.

(D) $SU(2)$ from bivectors (proved)

Define $J_k = \frac{1}{2}e_{ij}$ for (i, j, k) cyclic. These satisfy $[J_i, J_j] = \varepsilon_{ijk}J_k$ ($SU(2)$ Lie algebra) — standard result from the Clifford/spin connection. The 3 gauge 2-forms generate quantum spin.

(E) W_{ij} is grade-preserving (proved)

W_{ij} is Clifford grade-0

$$W_{ij} = \exp(-|\Omega_i - \Omega_j|^2/\varepsilon_0).$$

In $\text{Cl}(3, 0)$: for any vector $v = \sum_k v_k e_k$,

$$v^2 = \sum_k v_k^2 \text{ (scalar)}$$

because cross terms cancel by anticommutativity.

Therefore $|\Omega_i - \Omega_j|^2$ is a grade-0 (scalar) Clifford element, and W_{ij} is an exponential of a scalar — also grade-0.

A grade-0 coupling does not mix grades: if node i is in a grade- k mode, W_{ij} cannot induce transitions to grade- $l \neq k$ modes. **Grade is a conserved quantum number under W_{ij} coupling.**

(F) Uniform CRF ground state = Clifford null state (proved)

Define the Clifford norm of a probability distribution P :

$$C(P) = \sum_i P_i \cdot \text{sign}(e_i^2) = \sum_{k=0,1} (\text{prob. in grade } k) - \sum_{k=2,3} (\text{prob. in grade } k)$$

The grade-sign pattern for $\text{Cl}(3, 0)$: grades 0, 1 have $e^2 = +1$; grades 2, 3 have $e^2 = -1$. Counts: $1 + 3 = 4$ positive, $3 + 1 = 4$ negative (equal).

For the CRF ground state $P_i = 1/8$ (uniform):

$$C(P_{\text{gs}}) = \frac{4 \times (+1) + 4 \times (-1)}{8} = 0$$

The CRF ground state (maximum entropy) is the Clifford null state. Departure from uniformity (instability ε_0) creates non-zero $C(P)$.

(G) XOR structure = Clifford product index (structural)

The 8 modes are labeled by binary strings $b_2 b_1 b_0 \in \{0, 1\}^3$. The Clifford product index rule:

$$e_A \cdot e_B \text{ has index } A \oplus B \text{ (XOR)}$$

The XOR of two mode labels gives the index of their Clifford product. This is exact (the sign factor requires additional computation; the index rule is exact).

What Is Still Open

Programme for CRF Paper 2 (Clifford Dynamics)

1. **Born rule and Clifford multiplication.** Does $P_i \sim \text{Dir}(\alpha P_{\text{avg}})$ implement Clifford multiplication in some representation? Does P_{avg} transform as a Clifford element?
2. **Grade sectors and the 5 Psi-clusters.** The 5 Psi-clusters (V18) should correspond to the 5 grade sectors: grade-0, grade-2a, grade-2b, grade-2c, grade-3. Test: do different clusters have characteristic Ω -space patterns (scalar: $|\Omega| \approx 0$; gauge: Ω in planes; pseudo: $|\Omega|$ large)?
3. **Spinors as CRF states.** Minimal left ideals of $\text{Cl}(3)$ have dimension $n/2 = 4$ (real). Are there CRF node states of 4 real dimensions that naturally arise in simulation?

4. ε_0 **and the Clifford norm.** Is there a formula relating the CRF instability floor ε_0 to the Clifford norm $C(P)$ for typical non-ground states?
5. **Grade-Omega correspondence.** Predict and verify: grade- k modes occupy characteristic k -dimensional subspaces of Ω -space.

Status Table

Claim	Status	Basis
Grade structure $1 + 3 + 3 + 1 = 8$	Proved	exact count
$I_3^2 = -1$	Proved	Clifford axioms + Pauli matrices
I_3 in center (U(1) symmetry)	Proved	matrix algebra
SU(2) from bivectors	Proved	standard Clifford result
W_{ij} is grade-0 (grade-preserving)	Proved	$v^2 = v ^2$ in Cl(3,0)
Uniform state = Clifford null	Proved	grade count balance
XOR = Clifford product index	Proved	binary labeling
5 Psi-clusters = 5 grade sectors	Hypothesis	structurally motivated
Born rule implements Cl multiplication	Open	Paper 2
Spinors as CRF states	Open	Paper 2
ε_0 related to Clifford norm	Open	Paper 2

Summary

Seven independent features establish that the CRF $n = 8$ mode space is functionally Cl(3,0):

Proved: grade structure, $I_3^2 = -1$, U(1) center, SU(2) from bivectors, W_{ij} grade-preserving, ground state = null state, XOR product index.

The connection is not a coincidence of counting. It extends into the dynamics: W_{ij} (the fundamental coupling) is grade-0 and cannot mix grades. Grade is an approximately conserved quantum number in CRF.

The CRF ground state (uniform distribution, maximum entropy) is the Clifford null state ($C(P) = 0$). Instability ($\varepsilon_0 > 0$) corresponds to grade imbalance. The pseudoscalar mode $\{xyz\}$ provides the imaginary unit $I_3^2 = -1$ enabling complex

quantum mechanics.

The remaining question — what Clifford multiplication means for Born-rule dynamics — is the programme for Paper 2. The universe does not merely have the shape of $\text{Cl}(3)$.

It may actually compute in $\text{Cl}(3)$.